

Project Proposal – Societal Impact Modeling for AI Data Center Siting in Canada

What is the problem?

AI data centers are rapidly expanding across Canada. The country currently hosts 10.3 GW of data center capacity, with an additional 9 GW in the development pipeline — nearly doubling the installed base within the next few years. These facilities consume enormous amounts of electricity and water, and their environmental impact varies dramatically depending on where they are built.

A 100 MW data center in Alberta produces approximately 430,000 tonnes of CO₂ per year under standard grid-average accounting, while the same facility in Quebec produces roughly 1,500 tonnes — a 286-fold difference. But this comparison understates the real complexity. Research in the *Journal of Industrial Ecology* (Dandres et al. 2016) shows that the marginal electricity serving new data center loads in Canada is dominated by natural gas, with a GHG intensity of 0.85–1.01 kg CO₂-eq/kWh — roughly 5x higher than the Canadian average. Worse, 60–70% of this marginal electricity comes from reduced exports to the United States, triggering compensating fossil fuel generation south of the border.

Yet no publicly available tool exists to help policymakers evaluate these tradeoffs — including the hidden marginal and cross-border effects — before a siting decision is made. This gap is becoming urgent. In December 2025, Ontario passed Bill 40, the first Canadian law requiring ministerial approval for data center grid connections — a signal that governments are actively seeking analytical frameworks for these decisions. Meanwhile, data center siting choices lock in 20-30 years of environmental consequences. The decisions made today will shape Canada's carbon and water footprint for decades.

We propose building the first open-source Societal Impact Score for AI data center siting in Canada: a composite metric that evaluates carbon emissions, water consumption, grid stress, and cooling efficiency for any proposed location, with uncertainty quantification to support honest, evidence-based decision-making.

Why does this matter to our team?

We use AI every single day. ChatGPT to debug code at midnight before a deadline. GitHub Copilot to get unstuck. Cloud platforms to run experiments that our laptops can't handle. These tools feel effortless. But somewhere in the past year, studying systems and sustainability, we started asking ourselves: where does all of that come from?

What we found surprised us. Quebec runs on hydropower, so a data center there sounds “clean”. But the Dandres et al. research shows that the marginal electricity powering new Canadian data centers has a carbon intensity nearly six times the Canadian average. Why? Because Canada reduces its hydro exports to the US, and the US compensates by firing up coal and gas plants. The carbon is invisible under the accounting methods used.

Siting decisions lock in decades of environmental consequences. Ontario's Bill 40 now requires ministerial sign-off before a data center can connect to the grid and there's still no public tool to help those ministers make an informed decision. Our team brings together backgrounds in computer science, economics, and finance — a combination well suited to a problem that is as much about decision-making and policy as it is about modelling. We want to build something that matters, and we think an open tool that helps inform this process is exactly the kind of impact we came here hoping to have.

Why can machine learning help?

The environmental impact of a data center depends on nonlinear interactions between grid conditions, climate, water availability, and facility design. Machine learning is well suited to capturing these relationships.

Our approach involves four sub-models, each addressing a distinct prediction task:

- 1. Grid stress prediction:** A classification model trained on hourly electricity demand and reserve margin data to estimate whether adding a data center's load would push a provincial grid toward reliability limits.
- 2. Carbon intensity forecasting:** A time-series model forecasting both average and marginal carbon intensity of electricity at a given location. Marginal intensity — the carbon cost of the next unit of generation dispatched — is typically 2–5x higher than the grid average in mixed-source provinces (Dandres et al. 2016), and is the methodologically appropriate metric for evaluating new loads.
- 3. Cooling efficiency regression:** A regression model predicting Power Usage Effectiveness (PUE) — the energy overhead from cooling — based on local climate variables such as temperature and humidity.
- 4. Water intensity estimation:** A physics-based model estimating water consumption from cooling systems, weighted by local water stress indices.

These four outputs are combined into a single Societal Impact Score using configurable weights. Monte Carlo simulation propagates uncertainty from each sub-model through the composite score, producing confidence intervals rather than misleading point estimates.

What data do we plan to use?

We have identified three primary open datasets:

1. Environment and Climate Change Canada (ECCC) — Provincial Emission Factors

Official government emission intensities (gCO₂e/kWh) for every Canadian province and territory. These serve as the ground truth for our carbon intensity model.

(canada.ca/emission-factors)

2. Independent Electricity System Operator (IESO) — Ontario Hourly Generation Data

Hourly generation by fuel type (nuclear, hydro, gas, wind, solar), system demand, and market pricing. Available as CSV downloads and via the open-source gridstatus Python library.

(ieso.ca/power-data)

3. World Resources Institute (WRI) — Aqueduct 4.0 Water Risk Atlas

Sub-basin water stress indices covering all of Canada, including baseline water stress, seasonal variability, and drought risk. Licensed under Creative Commons Attribution 4.0.

(wri.org/aqueduct)

We also draw on the methodological findings of Dandres et al. (2016), published in the Journal of Industrial Ecology, which established that marginal generation in Canada is 85–100% fossil-fueled and identified cross-border electricity trade as a critical factor in consequential carbon accounting for new loads.

Does this project relate to coursework?

This project is purely extracurricular. It does not fulfill any coursework requirement. It is driven by our shared interest in applying machine learning to sustainability challenges.

Willingness to pivot

Our team is fully open to pivoting to an alternative community project if selected and our mentor recommends a different direction. We are here to learn and to contribute, and we trust the mentorship process.